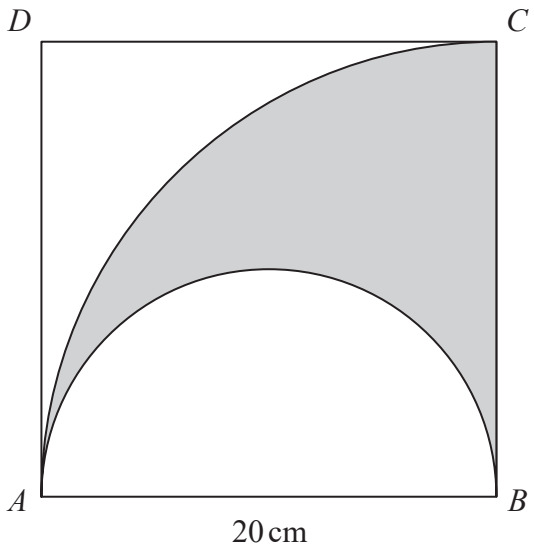
7 The diagram shows a square $ABCD$ with sides of length 20 cm.
It also shows a semicircle and an arc of a circle.



AB is the diameter of the semicircle.
AC is an arc of a circle with centre B.

Show that $\dfrac{\text{area of shaded region}}{\text{area of square}} = \dfrac{\pi}{8}$

Semicircle: $r = 10$; full circle area $= \pi \times 10^2 = 100\pi$

	Degrees	Area	
	360	100π	
÷2	180	50π	÷2

1 mark
180° is half of the full circle

Quarter circle: $r = 20$; full area $= \pi \times 20^2 = 400\pi$

	Degrees	Area	
	360	400π	
÷4	90	100π	÷4

1 mark
90° is a quarter of the full circle

Shaded area $= 100\pi - 50\pi = 50\pi$ cm^2

Square area $= 20 \times 20 = 400$ cm^2

1 mark
Subtract the areas; find the square area

$$\frac{\text{Shaded area}}{\text{Square area}} = \frac{50\pi}{400} = \frac{\pi}{8}$$

1 mark
Exact ratio shown; keep π

(Total for Question 7 is 4 marks)